

# 7-Day Study Plan: Python → Agentic AI → LangChain → LangGraph

Goal: Sep 7 – Sep 13 | Background: Strong in C++, MERN, SQL, MySQL, OTEL | New to Python | Target: working knowledge before 14 Sep

| Day / Date                        | Topics to Study                                                                                                                                                                                                                                                        | Video Sources                                                                                                                                                                     | Reading / Docs                                                                   |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Day 1</b><br><b>Sun 7 Sep</b>  | <b>Python basics for C++ devs:</b> syntax & indentation (no braces/semicolons), variables, data types, if/for/while, functions, lists/tuples/dicts ( $\approx$ vector/array/map), string formatting (f-strings), install Python + VS Code, venv & pip.                 | <ul style="list-style-type: none"><li>• 'Python for Programmers' – freeCodeCamp (YouTube)</li><li>• Bro Code 'Python Full Course' (YouTube)</li></ul>                             | W3Schools Python Tutorial (w3schools.com/python)                                 |
| <b>Day 2</b><br><b>Mon 8 Sep</b>  | <b>Python intermediate:</b> OOP (class, <code>__init__</code> , inheritance vs C++ classes), exception handling (try/except), list/dict comprehensions, <code>*args/**kwargs</code> , modules/imports, JSON handling, basics of async/await (needed later for agents). | <ul style="list-style-type: none"><li>• Corey Schafer – 'Python OOP Tutorials' (YouTube playlist)</li><li>• Corey Schafer – 'Python Async/Await' video</li></ul>                  | realpython.com (OOP + async articles)                                            |
| <b>Day 3</b><br><b>Tue 9 Sep</b>  | <b>Agentic AI concepts:</b> what is an LLM, tokens/embeddings/context window, prompt engineering basics, vector databases (concept only), what makes an "agent" vs a plain chatbot, the ReAct pattern (Reason+Act), tool/function calling.                             | <ul style="list-style-type: none"><li>• IBM Technology – 'What is Agentic AI?' (YouTube)</li><li>• IBM Technology – 'AI Agents vs Chatbots' / ReAct explained (YouTube)</li></ul> | docs.langchain.com/oss/python/langchain/overview (concepts page)                 |
| <b>Day 4</b><br><b>Wed 10 Sep</b> | <b>LangChain fundamentals:</b> chat models, prompt templates, output parsers, chains (LCEL), tools, basic memory, intro to RAG (retrieval-augmented generation). Write a small script: prompt → LLM → parsed answer.                                                   | <ul style="list-style-type: none"><li>• freeCodeCamp – 'LangChain Course for Beginners' (YouTube)</li><li>• LangChain official YouTube channel – Quickstart videos</li></ul>      | python.langchain.com — official Quickstart + LCEL guide                          |
| <b>Day 5</b><br><b>Thu 11 Sep</b> | <b>LangGraph fundamentals:</b> why LangGraph over plain LangChain (state machine for LLM control flow), StateGraph, State (like a struct), Nodes (Python functions), Edges & conditional edges, compiling & invoking a graph.                                          | <ul style="list-style-type: none"><li>• LangChain Academy – 'Introduction to LangGraph' (free course, video modules 1-2)</li></ul>                                                | docs.langchain.com/oss/python/langgraph/overview — Quickstart                    |
| <b>Day 6</b><br><b>Fri 12 Sep</b> | <b>LangGraph in depth:</b> checkpointing/memory across turns, human-in-the-loop (interrupts), building a ReAct tool-calling agent, basics of multi-agent graphs (supervisor pattern).                                                                                  | <ul style="list-style-type: none"><li>• LangChain Academy – modules 3-4 (Memory, Human-in-the-loop, Agents)</li></ul>                                                             | LangGraph 'Guides' section (docs.langchain.com) — streaming, memory, multi-agent |
| <b>Day 7</b><br><b>Sat 13 Sep</b> | <b>Build &amp; revise:</b> build one small end-to-end agent (e.g. a tool-using Q&A bot with 2-3 tools + memory) tying together Python + LangChain + LangGraph. Revise Day 1-6 notes; prepare questions for your Sr. Dev.                                               | <ul style="list-style-type: none"><li>• Follow official 'LangGraph Quickstart' tutorial hands-on (build along)</li></ul>                                                          | langchain-ai.github.io / GitHub 'langgraph' examples folder                      |

**How to use this:** Spend ~2-3 hrs/day. Watch the video first, then immediately type the code yourself (don't copy-paste) — this is the fastest way for a C++ dev to build Python muscle memory. Keep one running notes file per day (topic → 3-4 line summary → 1 code snippet you wrote). **Mental mapping tip:** Python *list*  $\approx$  C++ *vector*, *dict*  $\approx$  *map/unordered\_map*, *class* works like C++ classes but no explicit types/pointers, indentation replaces { }. **Extra (if time permits):** LangChain Academy (academy.langchain.com) is free and covers Days 4-6 in one structured track — use it as your spine and the YouTube videos as supplementary explanation.